

Project Overview

E-commerce is a rapidly evolving industry, and leveraging data analytics is key to staying competitive. In our case study, we demonstrate how Power BI can be utilized to extract valuable insights from e-commerce data, enabling businesses to make informed decisions and drive growth. From data preparation to visualization and sharing, Power BI offers a comprehensive solution for analyzing e-commerce data and gaining a competitive edge in the market.

Data Preparation:

- **Loading CSV file (5 Tables) :**

1. Customers
2. Orders
3. Order Details
4. Categories
5. Product

- **Cleaning and Transforming data to be prepared for modeling and analysis**

1. Check Data Types
2. Handling Outliers
3. Handling some nulls
4. Split Columns
5. Create Columns
6. Add Column (Category Id in Tables) To Make Relation By $\rightarrow =_xlfn.XLOOKUP(A2,Orders!B:B,Orders!L:L,0)$
7. Replace Values

Key Finding and Insights –According to Sales :

Regional Sales Performance

- Sales revenue was significantly high in the **USA and Germany**, with strong growth rates, contrasting with other countries like Poland and Norway where sales were comparatively lower.

Seasonal Trend

- Across the months, **April experienced the highest surge in sales**, followed by March, while June saw the lowest sales performance.

Key Finding and Insights –According to Orders Analysis:

Monthly Order Performance

– April witnessed the highest number of orders, followed by **March**, while **November** recorded the lowest.

Historical Order Peak

The highest number of orders occurred in the year **1997**.

Top Performing Location and Financial Return

– In the **USA**, the city of Rio de Janeiro stood out with a remarkable financial return of **\$267.87k**, accompanied by a conversion rate of **97.80%** and an average order value of **\$322.73**.

Category Analysis

The "**Fresh**" category registered the highest number of orders, totaling **354 orders**.

Key Finding and Insights –According to Customers Analysis:

Customer Distribution by Location

The highest number of customers was located in the USA, totaling 13.

Communication Channel Preference

The majority of customers engaged with the sales department, followed by the marketing department

–indicating a preference for direct sales interactions.

Order Placement by Department

Sales department interactions resulted in the highest number of orders, totaling 391 orders.

Key Finding and Insights –According to Products Analysis:

Product Portfolio Overview

The e-commerce platform offers a total of **77** products across various categories.

Category Distribution

The "Fresh" category boasts the largest number of products, with 12 items, followed closely by "**Sweets**" with **13** products and "Cheese" with 10 products
– Conversely, the "Fruits and Meats" category has the smallest number of products.

Top Product by Average Order Value (AOV):

The product "**Cote de Blaye**" achieved the highest average order value (AOV), indicating its premium status among customers.

Unit Sales Analysis:

The highest number of units sold was attributed to the "**Fresh**" category, while the "Fruits" category recorded the least number of units sold.

Recommendations



Recommendations: –Sales Analysis Recommendations:

Regional Focus: Given the high sales performance in the USA and Germany, consider allocating additional resources and marketing efforts to capitalize on these lucrative markets.

Targeted Marketing: Develop targeted marketing campaigns tailored to the preferences and behaviors of customers in regions with lower sales, such as Poland and Norway, to stimulate demand and increase sales.

Seasonal Promotions: Leverage the seasonal sales trends by offering promotions and discounts during peak months like April and March to maximize revenue and customer engagement.

Recommendations: –Orders Analysis Recommendations:

Customer Engagement: Strengthen the sales and marketing departments' capabilities to engage with customers effectively, as these channels drive the highest number of orders. Invest in training and tools to enhance customer relationship management.

Category Promotion: Given the popularity of the "Fresh" category, focus on promoting these products through targeted marketing campaigns and strategic placement on the e-commerce platform to drive further sales growth.

Location-based Strategies: Explore opportunities to replicate the success of Rio de Janeiro in other regions by analyzing local market dynamics and tailoring marketing strategies to resonate with the target audience.

Recommendations: –Customers Analysis Recommendations:

Customer Retention: Implement strategies to retain and nurture the existing customer base in the USA, such as personalized offers, loyalty programs, and excellent customer service, to foster long-term relationships and drive repeat purchases.

Multichannel Communication: Strengthen communication channels between customers and sales/marketing departments by integrating CRM systems and providing omnichannel support to enhance the customer experience and streamline interactions.

Sales Department Optimization: Further optimize the sales department's performance by providing additional training, refining sales processes, and incentivizing sales representatives to increase order conversion rates and revenue generation.

Dashboard



E-commerce Analysis



Orders



Sales



Products



Customers



By Ahmed Hamada

RESET

Year

All

Quarter

All

Month

All

Category

All

Product

All

Country

All

City

All

CustomerTitle

All

830

Total Orders

51K

Total Units Sold

62

Avg Units Per Order

\$1.27M

Total Sales

\$1,525

Avg Order Value

\$65K

Total Freight

Total Orders by Year, Quarter, Month and Weekday

88

83

103

105

46

30

55

58

60

64

59

79

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Total Orders by Category

fresh

354

cheese

303

sweets

295

sea

291

shampoos

193

breads

182

meats

161

fruits

129

Total Orders, Units, Sales by Countries

Country	Total Orders	Total Units Sold	Total Sales	Avg Units Per Order	Avg Order Value	Total Freight
USA	122	9,362	\$245,678.26	77	\$2,014	\$13,781
Germany	120	9,062	\$226,916.58	76	\$1,891	\$11,217
Brazil	84	4,296	\$108,789.18	51	\$1,295	\$4,892
France	76	3,227	\$80,918.32	42	\$1,065	\$4,205
UK	56	2,742	\$58,971.31	49	\$1,053	\$2,954
Venezuela	46	2,936	\$56,810.63	64	\$1,235	\$2,735
Austria	40	5,167	\$128,003.84	129	\$3,200	\$7,392
Sweden	37	2,235	\$54,495.14	60	\$1,473	\$3,238
Canada	30	1,984	\$50,196.29	66	\$1,673	\$2,198
Italy	28	822	\$15,770.15	29	\$563	\$864
Total	830	51,317	\$1,265,793.04	62	\$1,525	\$64,943

Note:

To see the **Orders** by years, quarters, months, and Weekdays please turn on the drill-down/up mode of the **Line Chart**.

Total Orders by Category

London

46

Rio de Janeiro

34

São Paulo

32

Boise

31

Graz

30

Cunewalde

28

México D.F.

28

Bräcke

19

Cork

19

Albuquerque

18

Luleå

18

San Cristóbal

18

RESET

Year

All

Quarter

All

Month

All

Category

All

Product

All

Country

All

City

All

CustomerTitle

All



8

Categories



77

Products



51K

Total Units Sold



3119

UnitsInStock



780

UnitsOnOrder



Orders



Sales



Products



Customers

Top 5 Products by Total Sales

Côte de Blaye	\$141.4K
Thüringer Rostbr...	\$80.4K
Raclette Courdav...	\$71.2K
Tarte au sucre	\$47.2K
Camembert Pierrot	\$46.8K

Top 5 Products by Total Orders

Raclette Courdav...	54
Camembert Pierrot	51
Tarte au sucre	48
Thüringer Rostbr...	32
Côte de Blaye	24

Top 5 Products by Total Units Sold

Camembert Pierrot	1,577
Raclette Courdav...	1,496
Tarte au sucre	1,083
Thüringer Rostbr...	746
Côte de Blaye	623

Bottom 5 Products by Total Sales

Chocolade	\$1.4K
Geitost	\$1.6K
Genen Shouyu	\$1.8K
Laughing Lumber...	\$2.4K
Longlife Tofu	\$2.4K

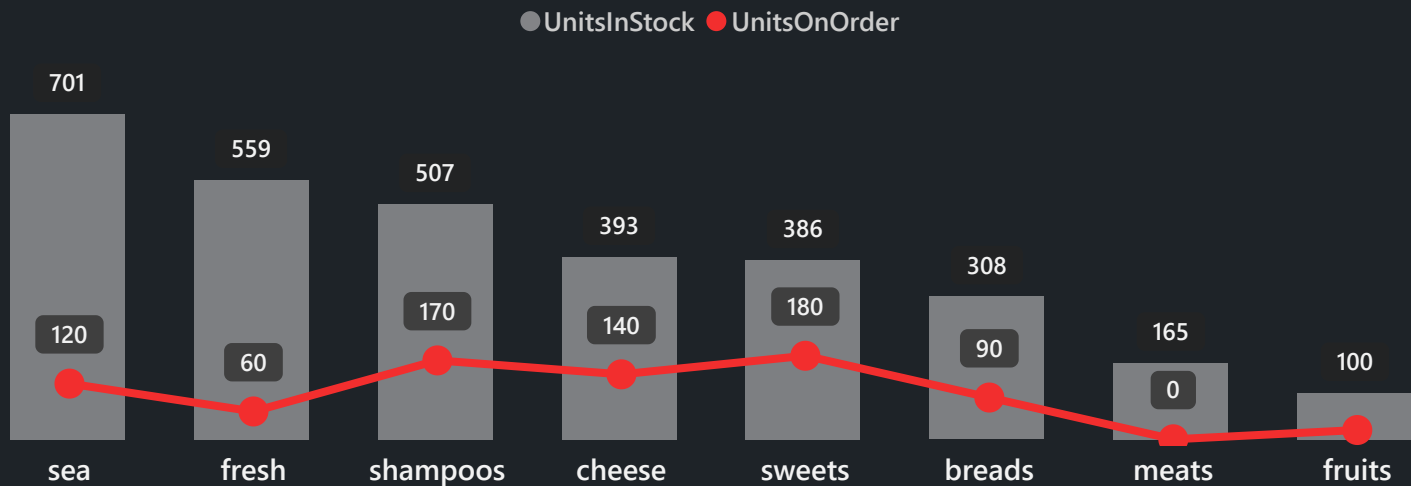
Bottom 5 Products by Total Orders

Chocolade	6
Genen Shouyu	6
Laughing Lumber...	10
Longlife Tofu	13
Geitost	32

Bottom 5 Products by Total Units Sold

Genen Shouyu	122
Chocolade	138
Laughing Lumber...	184
Longlife Tofu	297
Geitost	755

UnitsInStock and UnitsOnOrder by Category



Bottom 10 Products by Units In Stock

Alice Mutton	0
Chef Anton's Gumbo Mix	0
Gorgonzola Telino	0
Perth Pasties	0
Thüringer Rostbratwurst	0
Sir Rodney's Scones	3
Longlife Tofu	4
Louisiana Hot Spiced Okra	4
Røgede sild	5
Northwoods Cranberry S...	6
Scottish Longbreads	6



Year

Quarter

Month

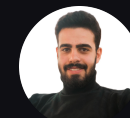
Category

Product

Country

City

CustomerTitle



91

Customers



89

Active Customers



21

Countries



69

Cities



Orders



Sales

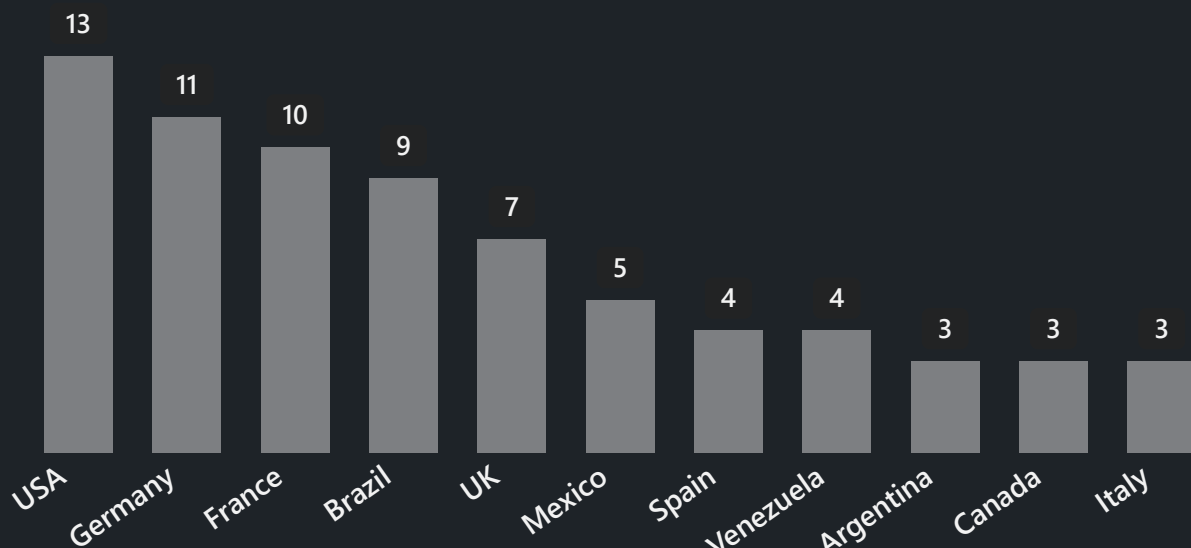


Products

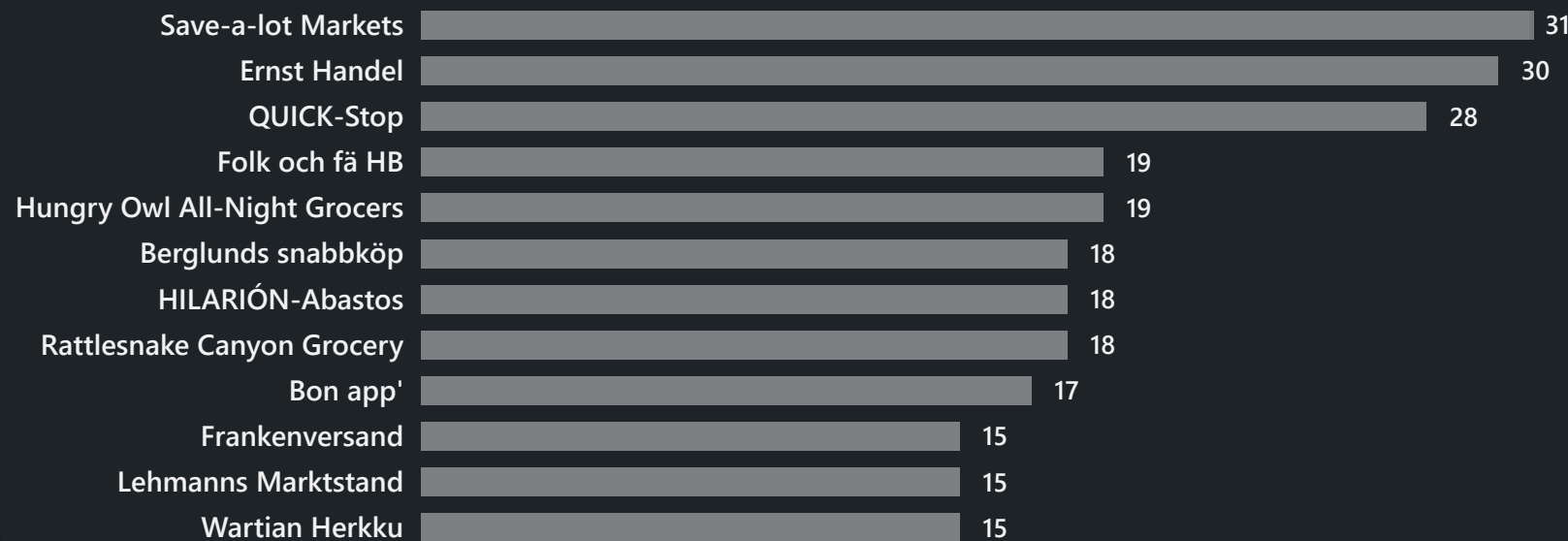


Customers

Top 10 Countries by Active Customers



Top 10 Companies by Total Orders



830

Total Orders

Top 10 Customers by Total Sales

Customer Name	CustomerID	CompanyName	CustomerTitle	Country	City	Address	Total Sales
Horst Kloss	QUICK	QUICK-Stop	Accounting Manager	Germany	Cunewalde	Taucherstraße 10	\$110,277.30
Roland Mendel	ERNSH	Ernst Handel	Sales Manager	Austria	Graz	Kirchgasse 6	\$104,874.98
Jose Pavarotti	SAVEA	Save-a-lot Markets	Sales Representative	USA	Boise	187 Suffolk Ln.	\$104,361.95
Paula Wilson	RATTC	Rattlesnake Canyon Grocery	Assistant Sales Representative	USA	Albuquerque	2817 Milton Dr.	\$51,097.80
Patricia McKenna	HUNGO	Hungry Owl All-Night Grocers	Sales Associate	Ireland	Cork	8 Johnstown Road	\$49,979.90
Mario Pontes	HANAR	Hanari Carnes	Accounting Manager	Brazil	Rio de Janeiro	Rua do Paço, 67	\$32,841.37
Philip Cramer	KOENE	Königlich Essen	Sales Associate	Germany	Brandenburg	Maubelstr. 90	\$30,908.38
Maria Larsson	FOLKO	Folk och få HB	Owner	Sweden	Bräcke	Åkergatan 24	\$29,567.56
Jean Fresnière	MEREP	Mère Paillarde	Marketing Assistant	Canada	Montréal	43 rue St. Laurent	\$28,872.19
Karl Jablonski	WHITC	White Clover Markets	Owner	USA	Seattle	305 - 14th Ave. S. Suite 3B	\$27,363.60
Total							\$570,145.05



Orders



Sales



Products



Customers



Data Analyst | Machine Learning Enthusiast

Strong Foundation in Data Processing, Statistical Analysis, and Data Visualization | Skilled in Data Analysis using Python, SQL, Power BI, and Excel | Proficient in Data Manipulation using Python | Passion for Uncovering Insights and Making Data-Driven Decisions | Skilled in Identifying Trends, Creating Reports, and Communicating Complex Findings Effectively | Fundamentals of Machine Learning (Regression and Classification Algorithms) using Scikit-Learn Framework.

